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SUPERSEDING
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MILITARY SPECIFICATION

BLOCKMAKING PLANT, CONCRETE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a concrete blockmaking plant. The plant shall be capable of manufacturing 8- by 8- by 16-inch nominal size blocks at a minimum rate of 1,500 blocks per day.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
TT-V-119	- Varnish, Spar, Phenolic-Resin.
MMM-A-260	- Adhesive, Water-Resistant (for Sealing Waterproofed Paper).
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3630

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MIL-V-173 - Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3600 - Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-C-3774 - Crates Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignition.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A532 - Abrasion-Resistant Cast Irons.
C90 - Hollow Load-Bearing Concrete Masonry Units.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 1 - Industrial Controls and Systems.
MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The concrete blockmaking plant shall consist essentially of a blockmaking machine, mixer, loader, water metering device, pallets, curing racks, and all controls necessary for manual and automatic operation. Unless otherwise specified (see 6.2), an off bearing hoist and a pallet return conveyor shall be furnished.

3.1.1 Standard commercial product. The concrete blockmaking plant shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the concrete blockmaking plant being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified (see 6.2), the contractor shall furnish a complete blockmaking plant for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.5 Design. The concrete blockmaking plant shall be designed to produce hollow concrete blocks that comply with the requirements of ASTM C90. The design shall be such as to facilitate maintenance, servicing, and cleaning. The plant shall be designed to prevent conditions hazardous to personnel or deleterious to equipment. Screw threads and pipe threads shall conform to the requirements of FED-STD-H28.

3.6 Performance. The concrete blockmaking plant shall meet the requirements specified herein. The mixer shall thoroughly and uniformly combine water, dry cement, and coarse and fine aggregates to produce the required moist concrete mix necessary for making concrete blocks. The mixer shall have a minimum capacity of 16 cubic feet (cu ft). The concrete loader shall have a minimum capacity of 16 cu ft, be automatic in operation, with a manual override switch that allows manual operation of the concrete loader. The conveyor hopper shall transport concrete from the mixer to the blockmaking plant when the applicable button is depressed. The hopper of the blockmaking machine shall have a capacity of at least 20 cu ft, and shall be removable. When specified (see 6.2), the blockmaking machine shall be furnished with a hopper extension to give a total hopper capacity of not less than 26 cu ft. When the cycle-start control is energized, the blockmaking machine shall automatically load the mold box from the feed drawer, drop the pressure head, and vibrate the concrete until the proper level is reached. An adjustable timing device shall regulate the operation of the feed drawer. The stripper mechanism shall remove the finished block from the mold. The finished blocks shall be transported to the offloading area on a pallet, by the rollout mechanism. The off bearing hoist shall move the pallet and blocks to the curing rack. The mold box shall be of a size adequate to make one 8- by 8- by 16-inch nominal size concrete block per cycle. The blockmaking machine shall make a minimum of 1,500 blocks per day. The blocks formed shall be free from voids and have smooth faces and sharp, square corners and edges. The blocks shall be uniform in size, shape, weight, color, texture, and finish. Performance test shall be verified in accordance with paragraph 4.4.1.

3.7 Construction.

3.7.1 Mixer. The batch mixer shall be the horizontal-drum type and shall be provided with mixing blades. The blades shall be adjustable and replaceable. The drum shall be fitted with abrasion-resistant, replaceable, sectional liners, and water distribution manifold. The liners and mixing blades shall be constructed of abrasion-resistant cast iron conforming to ASTM A532, type I, grade 1, or better. The drum charge opening shall be provided with a steel bar safety grill. The discharge door shall be manually operated or as specified hydraulically or pneumatically operated. The blade shaft shall be rotated at a speed of 20 to 25 revolutions per minute by means of an electric motor that drives a system of V-belts and precision gears or by direct gear motor drive. The motor shall be rated not less than 10 horsepower (hp). A lever-operated clutch shall be provided as a means of starting the fully-loaded mixer.

3.7.2 Conveyor loader. The conveyor loader shall be a belt conveyor of sufficient size to transport the concrete batch mix to the blockmaking machine.

3.7.3 Blockmaking machine. The blockmaking machine shall be of a plain pallet type. The blockmaking machine shall be provided with a vibrating system adequate to vibrate the material into the mold. The blockmaking machine shall

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also be provided with a feed drawer containing an agitating mechanism to loosen the material so that it will fall into the molds, a motorized conveyor, and a control panel. The machine shall be provided with a removable hopper. The frame shall be constructed of structural steel adequately braced to prevent vibration, distortion, and deflection. The blockmaking machine shall be set on rubber or other suitable pads as provided by the manufacturer, and shall not require a concrete pad to stabilize it. The feed drawer shall be mounted on rollers, and adequately supported to assure alinement. Each agitator shaft shall employ a replaceable shear pin or other safety device. The vibrator shafts shall be provided with a fixed directional vibration and shall be constructed to maintain adequate vibration on each mold used.

3.7.3.1 Mold assembly. The mold assembly shall be constructed of either welded steel or bolted type construction with removable liners, end plates, end cores, and core bar assemblies. The molds shall be made of an abrasion resistant steel. The mold assembly shall be supported in the machine to obtain effective vibration, and shall be readily removable from the machine. Electrical height pins shall be used for precision gaging of the height of the blocks produced.

3.7.4 Water metering device. The water metering device shall deliver a predetermined quantity of water to the mixer. The meter shall have an adjustable range of 3 to 30 gallons per minute. A manual tripping mechanism shall be provided for emergency shutoff.

3.8 Motors, motor starters, and motor overload protective devices. The motors shall be rated and constructed in accordance with the applicable provisions of NEMA MG 1, and shall be the squirrel-cage induction polyphase type. The motors shall be ball bearing, waterproof, and of adequate size to operate continuously at any ambient temperature between 40 degrees centigrade (°C) and 55°C, without overheating. Motors shall be rated not less than 120 percent of the actual load and provided with a motor starter and thermal overload protection. The motor starters and overload protection devices shall conform to the applicable requirements of NEMA ICS 1, and shall be housed in a waterproof master control box. The control box shall be either wall mounted or attached to the blockmaking machine with vibration-isolation mountings as specified (see 6.2).

3.9 Electrical wiring and components. The concrete blockmaking plant equipment shall be completely wired for operation when connected to the electrical source of supply. Wiring and electrical components such as controllers and switches shall conform to the applicable requirements of NFPA No. 70. All wire shall be color coded, suitably identified at all terminals and junctions, and enclosed in a rigid metal conduit. Liquid-tight flexible metal conduit may be used, as required, in accordance with NFPA No. 70. Tubing or conduit, fittings, junction boxes, and similar items shall be constructed of corrosion resistant metal or of metal with an integral corrosion resistant coating. A wiring diagram shall be furnished with each plant. Unless otherwise specified (see 6.2), the electrical components shall be designed for operation on a nominal 230 volt, 3 phase, 60 Hertz, alternating current system.

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3.10 Accessories. Accessories shall be as specified herein.

3.10.1 Mold boxes. A half-block mold box assembly or adapters shall be furnished to make a minimum of four standard 4- by 8- by 16-inch concrete blocks during each cycle. Unless otherwise specified (see 6.2), the contractor shall furnish the following:

a. One mold to make three blocks per cycle, blocks to be an exact size of 3-5/8 inches wide by 7-5/8 inches high by 15-5/8 inches long, with square ends, hollow type, three core, complete with pressheads.
b. One mold to make two blocks per cycle, blocks to be an exact size of 5-5/8 inches wide by 7-5/8 inches high by 15-5/8 inches long complete with pressheads.
c. One mold to make one block per cycle, blocks to be an exact size of 7-5/8 inches wide by 7-5/8 inches high by 15-5/8 inches long, complete with pressheads.
d. One 8-inch half block attachment consisting of corebar, pressheads, and shoes.
e. One 7-5/8- by 7-5/8- by 15-5/8-inch single corner attachment.
f. One 7-5/8- by 7-5/8- by 15-5/8-inch three core, single sash attachment.

3.10.2 Specialty molds. When specified (see 6.2), specialty molds shall be furnished in specified patterns.

3.10.3 Pallet. The pallet shall be the manufacturer's standard pallet constructed to fit the blockmaking machine and fabricated of 1/4-inch plain steel. The pallet shall be a minimum of 18- by 18-inches square ±1/16-inch to hold two 8- by 8- by 16-inch concrete blocks. The pallets shall be flat within 1/16 inch. Unless otherwise specified (see 6.2), the quantity of pallets to be supplied shall be 1,500.

3.10.4 Curing rack. The curing rack shall consist of three assembled steel sectional construction bays. Each rack shall store a minimum of 72 palletized 8- by 8- by 16-inch concrete blocks. The rack shall be approximately 70 inches long, 50 inches wide, and 65 inches high. The fully loaded rack shall be approximately 70 inches long, 50 inches wide, and 65 inches high. The fully loaded rack shall be capable of being transported by means of a forklift truck. Unless otherwise specified (see 6.2), forty 3 bay racks shall be furnished.

3.10.5 Pallet return. Unless otherwise specified (see 6.2), a motorized conveyor system shall be furnished to automatically convey empty pallets into the pallet receptacle. A minimum 3/4 hp motor shall drive a gear reduction unit to operate the conveyor.

3.10.6 Off bearing hoist. Unless otherwise specified (see 6.2), a swing hoist shall be furnished. The hoist shall be either electric, hydraulic, or pneumatic as specified (see 6.2). A magnetic lifting device and direct current rectifier shall be provided.

3.11 Hydraulic powered plant. When the blockmaking plant is powered by a hydraulic system, the hydraulic system shall be furnished complete with an electric-motor-driven, tank-mounted, hydraulic pump and all necessary controls, hoses, valves, and fittings.

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3.12 Pneumatic powered plant. When the plant is powered by a pneumatic system, a tank-mounted air compressor shall be equipped with electric motor, air receiver, constant speed unloader, pressure gage, safety valve, receiver drain valve, air filter, and all other required controls, hoses, valves, and fittings.

3.13 Lubrication. Lubrication shall be in accordance with the equipment manufacturer's recommendations. Pressure lubrication shall not damage seals or other parts. All parts requiring lubrication shall be lubricated prior to delivery. A lubrication chart (metal or decal, protected by clear plastic) shall be affixed to each machine, to indicate all points requiring lubrication, the grade of lubricant required, and the time interval for accomplishing lubrication.

3.14 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.15 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.16 Electromagnetic interference control. When specified (see 6.2), the equipment shall conform to the electromagnetic interference control requirements and test limit for class B equipment as specified in MIL-STD-461.

3.17 Identification marking. The concrete blockmaking plant, accessories, and replaceable parts shall be marked for identification in accordance with MIL-STD-130.

3.18 Instruction plates. The equipment shall be provided with instruction plates describing special or important procedures for operating and servicing equipment or warnings of hazardous procedures. The plates shall be conspicuously located. Brass screws or bolts not less than 1/8-inch in diameter shall be used to affix the plates to the equipment.

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3.19 Repair parts, service, and maintenance tools. Any special tools required for installation and maintenance shall be furnished with the plant.

3.20 Workmanship. The blockmaking plant shall conform to the quality and grade established by this specification.

3.20.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.20.2 Bolted connections. All boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided. All nuts, bolts, and screws shall be tight.

3.20.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet head shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

3.20.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.20.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one complete blockmaking plant when a first article sample is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be a standard production item from the contractor's current inventory provided the blockmaking plant meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining blockmaking plants to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on each unit. This inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each blockmaking plant shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Performance test. The first article shall be tested to verify compliance with the performance requirements of 3.6. The mixer shall be charged with sufficient fine and coarse aggregate and cement, and mixed with water to produce 30 cu ft of the desired moist concrete mix. All components shall operate, and automatic and manual control functions shall be performed during the test. The blockmaking machine shall manufacture three blocks per minute until the concrete is consumed. Failure of the plant to meet the performance requirements specified in 3.6, overheating of the plant, or other malfunction will be considered cause for rejection.

4.4.2 Electromagnetic interference control test. When conformance with electromagnetic interference limits is required, the first article shall be tested in accordance with MIL-STD-462 to determine conformance with electromagnetic interference control requirements of 3.16.

4.5 Quality conformance tests.

4.5.1 Functional test. Each concrete blockmaking plant shall be tested to verify compliance with the performance requirements of 3.6. The test shall be conducted for a period of not less than 1 hour. Any control, whether automatic or manual, component, or accessory which shows excessive vibration, overheating, or malfunction shall be rejected (see 4.4.1).

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4.5.2 Pallet inspection. Pallet samples selected shall be tested for flatness as specified in 3.10.3. The pallet shall be placed on a known level surface. A straight edge, at least 30 inches in length, shall be moved across the pallet. The test shall be performed on the reverse side of the pallet also. Failure to pass flatness requirements shall be cause for rejection.

4.5.2.1 Sampling plan for pallet inspections. Sampling shall be in accordance with MIL-STD-105. A lot shall consist of all pallets made of the same material and offered for delivery to the Government at one time under a specific contract. The inspection level shall be S-2 and the Acceptable Quality Level shall be 1.5 percent defective.

4.6 Packaging inspection. The packaging, packing, and marking of the complete concrete blockmaking plant shall be inspected to determine conformance with the applicable requirements of section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.5 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.6 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

5.1.1.7 Vibrator of blockmaking plant. The vibrator mechanism shall be secured to prevent vibration during shipping of the blockmaking plant.

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5.1.1.8 Material hopper. The material hopper of the blockmaking plant shall have the opening to the hopper sealed with barrier material as specified in PPP-B-1055, class H-5, L-4, or M-1.

5.1.1.9 Conveyor belts. Metal facings shall be coated with varnish conforming to TT-V-119 or MIL-V-13811. Each belt shall be completely rolled with a tube or piece of round wood placed in each end fold of the belt to prevent sharp bends. Prior to rolling, belt surfaces shall be dusted with powdered talc to prevent surfaces from adhering together. When required, to hold belt in compact roll, strapping conforming to QQ-S-781, class 1, type I, finish B, shall be placed around the roll. Suitable protective cushioning shall be placed under the strapping, and the strapping not tensioned so tight as to damage the belting. Each rolled belt shall be wrapped with waterproofed barrier conforming to PPP-B-1055, class E-1 or E-2, with all laps, seams, and folds sealed with adhesive conforming to MMM-A-260.

5.1.1.10 Portable conveyors. Portable conveyors shall be shipped knocked down to the most practicable extent. The structural components shall be nested, arranged, and secured with bolts, or steel strapping, in combination with suitable wood blocking or battens, as required to form nonshifting bundles. The strapping shall be 0.035 by 1-1/4 inch strapping conforming to QQ-S-781, class 1, type I, finish B. The strapping shall be stapled to any wood blocking or battens provided. Where it is not practicable to disassemble the equipment to the extent required for suitable bundling, the sections shall be secured to skid bases conforming to the skid type base of MIL-C-104, style A crates, and shipped as individual assemblies. The electric motors, conveyor belts, bolts, nuts, washers, and any other small components requiring the protection of a container for safe convenient handling shall be packed in containers as specified for disassembled components. Where skid bases are used, consideration shall be given to consolidating and securing any bundled or boxed components on the skid bases.

5.1.1.11 Molds. Molds shall be packaged method III.

5.1.1.12 Racks. Racks shall be packaged method III.

5.1.1.13 Pallets. Pallets shall be packaged method III.

5.1.1.14 Off bearing hoist. The hoist and cable shall be packaged method I, and lubricated for service.

5.1.1.15 Hydraulic system. When the blockmaking plant is powered by a hydraulic system, the hydraulic system shall be lubricated for service, and the reservoir tank for hydraulic fluid shall be lubricated as recommended by the manufacturer.

5.1.1.16 Pneumatic system. When the blockmaking plant is powered by a pneumatic system, the pneumatic system shall be preserved in accordance with the level A requirements of MIL-C-3600.

5.1.2 Level C. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The complete blockmaking plant shall be packed in a crate conforming to MIL-C-104, type II. Contents shall be cushioned, blocked, braced, and waterproofed, in accordance with MIL-STD-1186.

5.2.2 Level B. The complete blockmaking plant shall be packed in a crate conforming to MIL-C-3774, type II. Contents shall be cushioned, blocked, braced, and waterproofed in accordance with MIL-STD-1186.

5.2.3 Level C. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The concrete blockmaking plant is intended to be used for the production of hollow concrete blocks for masonry construction at advanced bases.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If specifications, standards, and handbooks are to be other than as listed in DoDISS (see 2.1.1).
c. When an off bearing hoist and a pallet return conveyor are not required (see 3.1, 3.10.5, and 3.10.6).
d. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
e. When a hopper extension shall be furnished (see 3.6).
f. Weather control box shall be wall mounted or attached to the blockmaking machine (see 3.8).
g. When electrical components shall be designed for operation on a power supply other than that specified (see 3.9).
h. When other than molds specified are to be provided (see 3.10.1).
i. When specialty molds are required and the pattern(s) required (see 3.10.2).
j. The quantity of pallets if other than specified (see 3.10.3).
k. The number of curing racks if other than specified (see 3.10.4).
l. When a motorized conveyor system for pallet return shall be furnished (see 3.10.5).
m. Whether off bearing hoist shall be pneumatic, electric, or hydraulic (see 3.10.6).
n. Cleaning, treatment, painting, and color of finish coat if other than specified (see 3.14).
o. When fungus treatment is required (see 3.15).
p. When electromagnetic interference control is required (see 3.16).
q. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition

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Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 3.2 and 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

Custodian:

Navy - YD

Preparing activity:

Navy - YD

(Project 3630-N003)

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